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Data Science

Wild Track

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Project Guide
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1. Introduction

Problem Identified

- Tracking the real-time movement of wild animals is difficult, which increases the risk of unexpected encounters and safety issues for people living near forest areas.
- There is no proper digital system for reporting, identifying, and monitoring unregistered or newly spotted wild animals, making it harder for authorities to respond quickly.

Solution Proposed

- Develop a mobile application that uses GPS and map integration to provide real-time tracking of wild animals, helping users and authorities stay informed.
- Introduce a feature that allows users to capture and upload images of unidentified animals along with location data, enabling the forest department to take timely action.

2. Objectives

- To develop a smart wildlife tracking system using modern digital technologies.
- To provide real-time location tracking of wild animals.
- To enhance safety for local residents through instant alerts.
- To assist forest departments in monitoring and managing wildlife efficiently.
- To support students and researchers with accurate wildlife data.

3. Scope

- Can help forest departments monitor animal movements effectively.
- Can improve safety for people living near forest areas.
- Can be used by students for wildlife research and studies.
- Can be expanded to national parks and wildlife reserves.
- Can enable reporting and tracking of unidentified animals.

4. Feature /Functionality

Feature 1: Secure Login

User authentication for authorized access (Forest officials, researchers, public users).

Feature 2: Live Animal Tracking Map

Displays real-time locations of tracked animals on an interactive map.

Feature 3: Nearby Animal Alerts

Sends notifications when a wild animal is detected near the user’s location.

Feature 4: Report Unknown Animal

Users can click photos of unidentified animals and upload them with location details to notify the forest department.

Feature 5: Data for Research

Provides historical movement data useful for students and wildlife researchers.

5. Outcome of Project

- Users can view real-time locations of wild animals on an interactive map to stay aware and avoid dangerous areas.
- Users receive instant alerts when wild animals are nearby, improving safety for people living near forest regions.
- Forest departments can efficiently monitor animal movements using real-time data for better wildlife management.
- Users can report unidentified animals with images and location details to help authorities take quick action.
- The system helps reduce human-wildlife conflicts by providing timely information and early warnings.
- The app supports students and researchers by providing useful data on animal movement and behavior.

6. Technology Stack

Front End (GUI):-

- **Framework/Language:** Flutter (Dart) for cross-platform Android.
- **State Management:** Provider for real-time data and auth states.
- **UI:** Custom map for tracking, Google Fonts (Poppins), Material Design 3.

Back End (Database & Services):-

- **Database:** Cloud Firestore (NoSQL) with real-time listeners.
- **Messaging:** Notification for alerts and confirmations.

7. Block Diagram

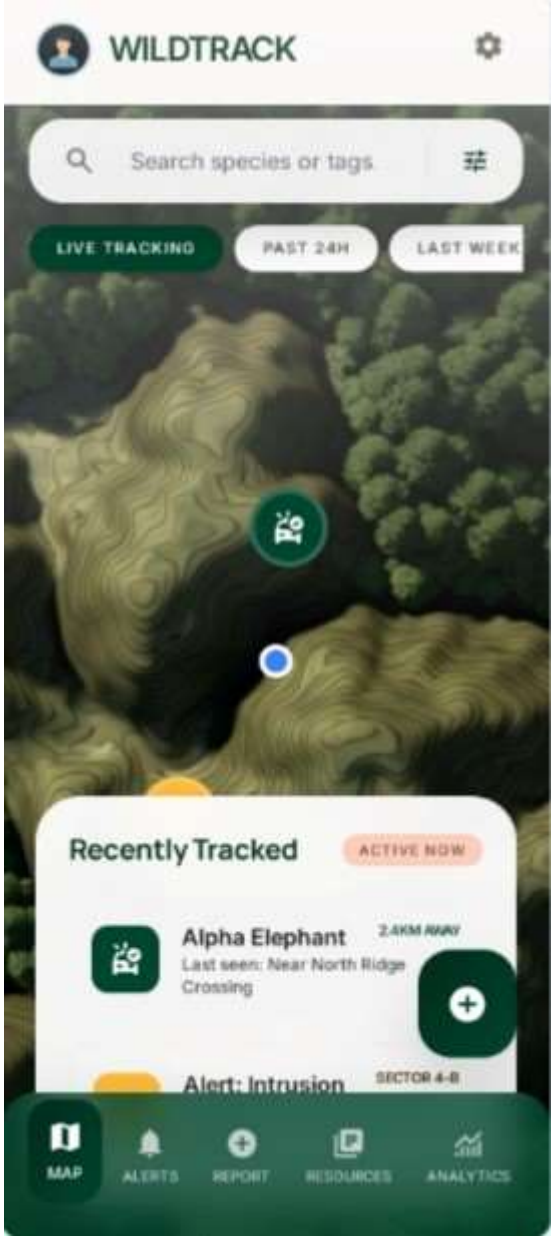
SDG

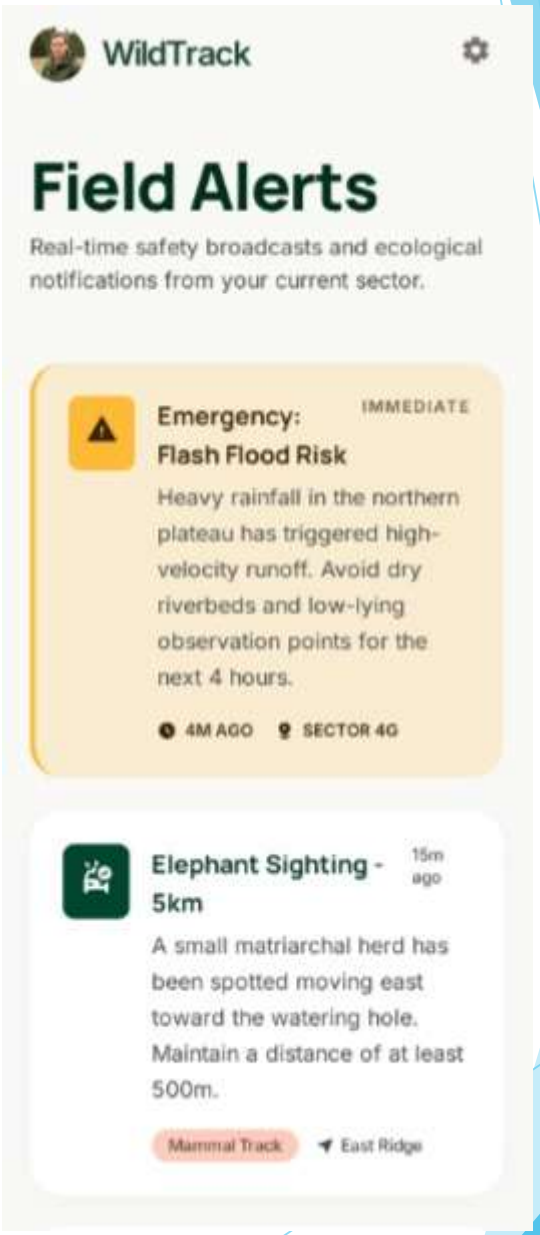
SDG 3: Good Health and Well-being

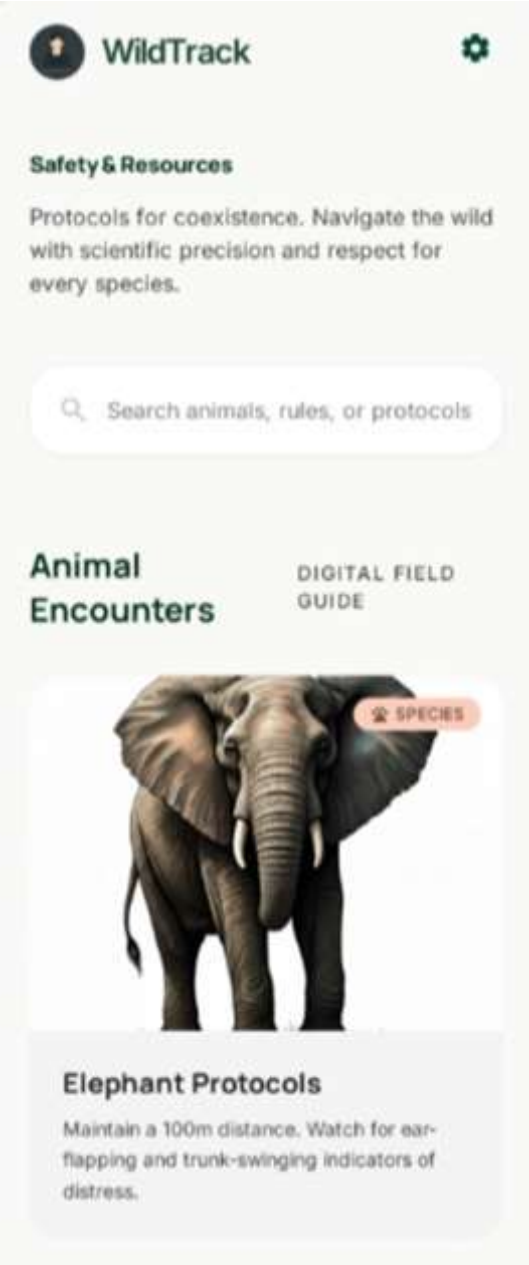
- Reduces risk of human-wildlife conflict by providing time alert about nearby wild animals
- Improves safety and awareness for people living ear forest
- Helps create a safer and more secure environment for both human and wildlife

SDG 9: Industry, Innovation, and Infrastructure

Project uses modern technologies like mobile apps, real-time databases, and digital systems to improve infrastructure within the campus. It promotes **innovation** by introducing a smart solution for parking management


WILDTRACK
Search species or tags
LIVE TRACKING
PAST 24H
LAST WEEK
Recently Tracked
ACTIVE NOW
Alpha Elephant
2.4KM AWAY
Last seen: Near North Ridge Crossing
Alert: Intrusion
SECTOR 4-B
MAP
ALERTS
REPORT
RESOURCES
ANALYTICS


WildTrack
Field Alerts
Real-time safety broadcasts and ecological notifications from your current sector.
Emergency: Flash Flood Risk
IMMEDIATE
Heavy rainfall in the northern plateau has triggered high-velocity runoff. Avoid dry riverbeds and low-lying observation points for the next 4 hours.
4M AGO
SECTOR 4G
Elephant Sighting - 5km
15m ago
A small matriarchal herd has been spotted moving east toward the watering hole. Maintain a distance of at least 500m.
Mammal Track
East Ridge


WildTrack
Safety & Resources
Protocols for coexistence. Navigate the wild with scientific precision and respect for every species.
Search animals, rules, or protocols
Animal Encounters
DIGITAL FIELD GUIDE
SPECIES
Elephant Protocols
Maintain a 100m distance. Watch for ear-flapping and trunk-swinging indicators of distress.

WildTrack
Analytics Dashboard
Real-time ecological movement and conflict monitoring
TOTAL SIGHTINGS (LAST 24H)
1,284
+12% from yesterday
ACTIVE CONFLICT ALERTS
08
3 high priority zones
RESOLVED INCIDENTS
142
94% resolution rate

Thank You…!!